University of Windsor – Master of Management
BSMM.8730 Data Acquisition and Management | Dr. Andreas S. Maniatis | Fall 2026

D.3.1 Midterm Project Status Report

Charging the Gap: EV Adoption vs. Public Charging Infrastructure in Ontario

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Introduction

Our project helps municipal planners and charging network operators identify which Ontario regions have electric vehicle (EV) registrations growing faster than public charging capacity. The solution combines structured CSV files, a semi-structured JSON API and unstructured web text in Python, stores the integrated data in a MySQL database, and will present the results in a Power BI dashboard. This report summarizes the progress made since the proposal was approved, the evidence available in our GitHub repository and Jira board, and the work that remains before the final deliverables on December 1.

1. Project Progress Breakdown

1.1 Task Breakdown Structure

The project was broken down into 21 tasks that follow the phases of the Data Analytics Lifecycle and are scheduled in two-week Scrum sprints in Jira. Table 1 lists every task with its dependencies and current status. Fourteen of the 21 tasks are completed, one is in progress and six remain.

ID	Task / milestone	Depends on	Status	Remarks / evidence
T1	Topic selection, problem statement and D.1 pre-proposal	–	Completed	Submitted Sep 27; instructor feedback received
T2	Updated D.1: three data types, web scraping, MySQL	T1	Completed	Resubmitted Oct 2 following feedback
T3	Jira Scrum board and GitHub repository set-up	T1	Completed	Backlog, six sprints, folder structure
T4	Acquire charging stations from NLR/NRCan API (JSON)	T3	Completed	4,608 Ontario stations, raw JSON stored
T5	Download quarterly EV-by-FSA files (Ontario Data Catalogue)	T3	Completed	All published quarters to Q2 2026
T6	Download StatCan registrations, FSA population and boundaries	T3	Completed	Tables 20-10-0025-01, 98-10-0019-01; 92-179-X
T7	Web scraper 1: catalogue page → quarterly CSV links	T5	Completed	BeautifulSoup; links saved to CSV
T8	Web scraper 2: Government of Canada news releases	T3	In Progress	Search page parsing being fixed; manual URL list as fallback
T9	Flatten JSON, clean and standardize, derive FSA	T4	Completed	Null ports → 0, duplicates removed
T10	Append EV quarters, join population, compute metrics	T5, T6, T9	Completed	EVs per port, ports per 1,000, gap score
T11	MySQL schema design (5 tables, keys, v_fsa_gap view)	T10	Completed	01_create_schema.sql
T12	SQL dump, import in MySQL Workbench, verification queries	T11	Completed	ev_charging_gap populated and checked
T13	D.4.1 User Requirements and Analysis document	T2	Completed	31 functional, 15 non-functional requirements
T14	D.4.2 Analysis and Design document	T11, T13	Completed	Submitted Oct 30

ID	Task / milestone	Depends on	Status	Remarks / evidence
T15	Text extraction into funding_announcements table	T8	Remaining	Regex rules written; awaits T8 output
T16	Spatial join for stations without a valid postal code	T6, T9	Remaining	Sprint 4
T17	Power BI data model connected to MySQL	T12	Remaining	Starts this sprint; read-only MySQL user
T18	Dashboard: map, gap ranking, trends, filters	T17	Remaining	Sprint 4
T19	EV projection (linear trend, four quarters)	T10	Remaining	Sprint 4
T20	Code comments, README, D.4.3–D.4.5, D.5 report	T15–T19	Remaining	Due Dec 1
T21	D.6 in-class presentation	T20	Remaining	Dec 9

1.2 Sequencing and Progress Since Initiation

The tasks were sequenced so that each phase produces the input of the next. Discovery tasks (T1–T3) defined the problem, the stakeholders and the tools, and set up Jira and GitHub before any data work started. Data acquisition (T4–T8) followed, because cleaning and modelling cannot begin until the sources are available. Preparation and integration (T9–T10) depend on all acquisition tasks except the news-release scraper, which is why the database could be built while the scraper is still being refined. The MySQL schema (T11) was designed only after the cleaned tables and their keys were known, and the database import (T12) is now the foundation for the remaining analysis and dashboard tasks (T15–T21). The requirements and design documents (T13–T14) were written alongside the build so that the remaining work is guided by agreed, testable requirements.

Since project initiation the team has moved from an idea to a working data foundation. All four primary datasets have been acquired from official sources, including 4,608 charging stations retrieved through the API, every quarterly EV-by-FSA file published to Q2 2026, and the Statistics Canada registration, population and boundary files. A Python notebook now flattens, cleans and integrates these sources, calculates the core metrics, and generates a self-contained SQL dump that loads the ev_charging_gap database in MySQL Workbench. The main open item in acquisition is the news-release scraper: the Government of Canada search page did not return results in the expected structure, so the parsing logic is being adjusted and a manually collected list of release URLs is being prepared as a fallback. The next sprint focuses on connecting Power BI to MySQL and building the first dashboard pages, which will turn the current database prototype into the visual prototype of the final application.

2. GitHub Repository and Version Control Evidence

All project work is tracked in the GitHub repository [GitHub repository link], and tasks are managed on the Jira board [Jira board link]. The repository is organized as the project portal: data/raw holds the original files exactly as downloaded, data/processed holds the cleaned outputs of the notebooks, notebooks holds the Python acquisition and wrangling code, sql holds the schema, dump and verification scripts, dashboard will hold the Power BI file, and docs holds the proposal, the requirements and design documents and the screenshots used as evidence. A main README describes the project, the team and the deliverables, and a data README records the source, link, download date and selection of every dataset, which provides traceability for each file. Personal credentials such as the API key are never committed.

Each member commits under their own GitHub account with messages that describe the actual change, such as adding the quarterly EV-by-FSA files, adding the JSON flattening and FSA derivation step, or adding the MySQL schema and verification queries. Commit messages reference the related Jira task where possible, so the history can be followed from requirement to implementation.

Nidhi Tandel, as Project Manager and Data Engineer, set up the repository and Jira board, acquired the API and CSV data, developed the acquisition and wrangling notebook, designed the MySQL schema, produced

and imported the SQL dump, and led the D.4.1 requirements document. Temitope Lawal, as Data Analyst and BI Developer, validated the downloaded datasets, defined the cleaning rules and the data dictionary, contributed to the D.4.2 design document, and leads the Power BI model and dashboard that start in the next sprint. Shiv [Last Name], who rejoined the team in October, was brought up to date on the full pipeline, contributed to the D.4.1 and D.4.2 documents, and is assigned to completing the news-release scraper and the EV projection module.

3. Demonstration of Completed Work

The completed tasks in Table 1 will be shown in a live walkthrough, in the same order as the pipeline. First, the GitHub repository will be opened to show its folder structure, the data README and the commit history, together with the Jira board showing the completed and remaining tasks in each sprint. Second, the acquisition and wrangling notebook will be run in Google Colab: it calls the NLR/NRCan API and receives the JSON for 4,608 Ontario stations, scrapes the Ontario Data Catalogue page for the quarterly CSV links, loads the EV, population and registration files, and prints the data-quality checks, including the number of stations without a valid postal code and the FSAs that have EVs but no public charging port.

Third, the MySQL database will be shown in Workbench. The schema contains five related tables – fsa, ev_registrations, charging_stations, vehicle_registrations_prov and funding_announcements – with primary and foreign keys, and the v_fsa_gap view that calculates EVs per public port and ports per 1,000 residents for every FSA and quarter (Figure 1). The verification script returns the row count of every table (Figure 2), the fifteen FSAs with the highest number of EVs per public port in the latest quarter (Figure 3), and the split between Level 2 and DC fast ports by region. Finally, the entity-relationship diagram generated by reverse engineering the database will be shown to demonstrate that the implemented schema matches the D.4.2 design.

[Insert Figure 1 – Workbench schema tree; Figure 2 – row counts; Figure 3 – top 15 gap query; Figure 4 – ERD, each reduced to about one-third of the page width]

These components show that the data acquisition, preparation and storage layers of the application are working and verified. The remaining work is concentrated in the presentation layer and the unstructured-data module, and the team is on track to deliver the complete application, documentation and dashboard by December 1.